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Inhibition of Cyclooxygenase 2 in Mice Increases Production of G-CSF and Induces Radioprotection

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Meloxicam, a selective inhibitor of cyclooxygenase 2, was tested to determine its ability to modulate hematopoiesis and to influence survival of mid-lethally γ -irradiated mice. A single dose of meloxicam (20 mg/kg) administered to mice intraperitoneally 1 h before irradiation was shown to enhance serum levels of granulocyte colony-stimulating factor (G-CSF) during the first 24 h after irradiation, to elevate numbers of granulocytic precursor cells in bone marrow and granulocyte counts in peripheral blood on day 10 after irradiation, and to increase 30-day survival of these mice. The results provide new evidence for the protective ability of meloxicam administration to mice irradiated with mid-lethal doses and contribute to the understanding of the mechanisms of this meloxicam action by drawing attention to the possible role of increased endogenous G-CSF production. © 2008 by Radiation Research Society

INTRODUCTION

Identifying drugs that meet the requirements for protection against and/or treatment of radiation-induced damage is currently one of the most important tasks in radiobiology (1, 2).

Classical non-steroidal anti-inflammatory drugs (NSAIDs) like indomethacin and diclofenac have been found to support recovery from myelosuppression induced by sublethal radiation doses [(3–8), reviewed in (9)]. These drugs act as inhibitors of prostaglandin production, and their protective effects have been shown to result in the removal of negative feedback control of myelopoiesis mediated by prostaglandins of the E series (10–12). One disadvantage of these drugs is their non-selective action on two forms of cyclooxygenases (COX) that catalyze the for-

mation of prostaglandins, i.e. the constitutive COX1 and the inductive COX2. This feature of the classical NSAIDs is reflected in the fact that, in addition to supporting myelopoietic recovery, they also produce undesirable effects connected with inhibition of COX1 such as gastrointestinal ulcerogenic activity. These effects have been reported especially at higher radiation doses that induce both the bone marrow syndrome and gastrointestinal damage; both histologically demonstrable damage to the intestinal wall and impaired survival have been observed for COX inhibitors administered before (13) or after irradiation (14).

We therefore examined the effect of meloxicam, an NSAID that preferentially inhibits COX2, possesses a pharmacodynamic and pharmacokinetic profile that appears to differ from that of the classical NSAIDs, and has a very high therapeutic index (15). In our recent paper (16), we described the ability of meloxicam administered to mice exposed to sublethal radiation doses to enhance hematopoietic reconstitution. The aim of the present study was to investigate the potential of meloxicam to influence survival of mice irradiated with mid-lethal doses of γ radiation. These studies were supplemented by additional experiments evaluating the state of the bone marrow hematopoiesis in the critical period preceding manifestations of lethality. Attention was also focused on the important hematopoietic cytokine G-CSF and on the role of meloxicam in modulating its serum levels.

MATERIALS AND METHODS

Animals

B10CBaF₁ male mice aged 3 months and weighing an average of 30 g were obtained from the breeding facility of the Medical Faculty, Masaryk University, Brno, Czech Republic. A total of 215 mice were used in this study. The mice were kept under controlled conditions; standardized pelleted diet and HCl-treated tap water were available *ad libitum*. The use and treatment of the animals followed the European Community Guidelines as accepted principles for the use of experimental animals. The experiments were performed with the approval of the Institute's Ethical Committee.

Meloxicam and its Administration

Meloxicam (Sigma, St. Louis, MO) was dissolved in sterile saline and administered intraperitoneally at a dose of 20 mg/kg in a volume of 0.2

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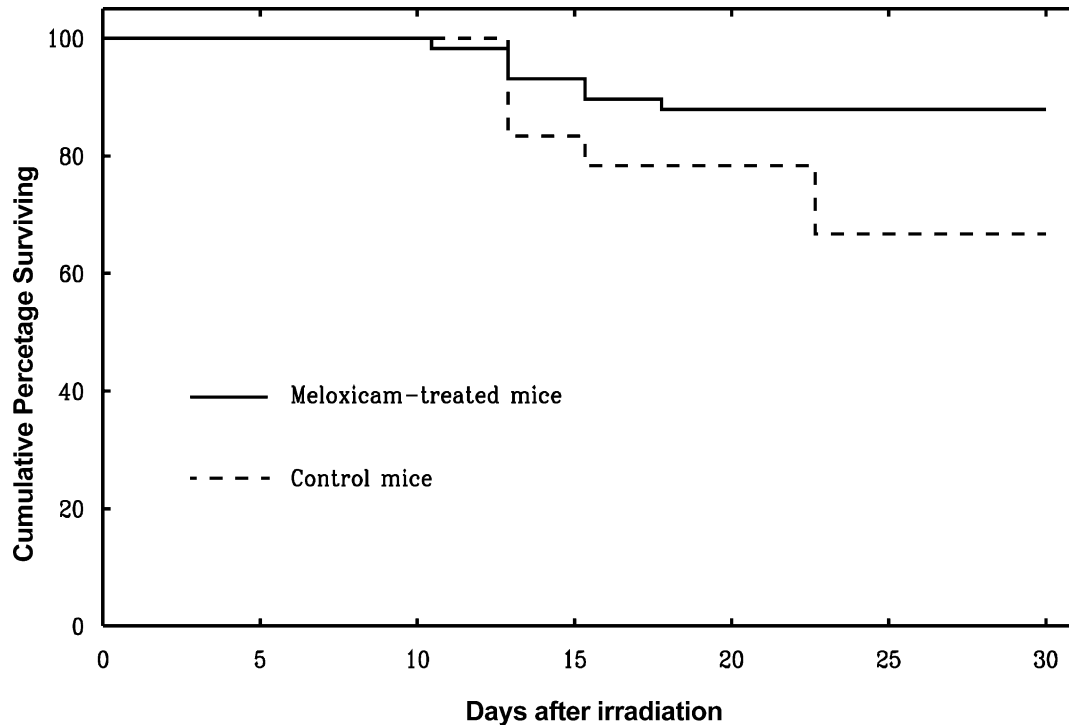


FIG. 1. Mice were given a single dose of meloxicam 1 h before irradiation with 7.50 Gy γ rays. Meloxicam-treated mice, $n = 48$; saline-treated mice, $n = 50$. Statistical significance: meloxicam-treated mice compared to saline-treated mice, $P < 0.01$ (Peto & Peto Wilcoxon test).

ml. The drug was always given as a single dose to nonirradiated mice or 1 h before irradiation. Sterile saline was used for control injections.

Irradiation

The mice were whole-body-irradiated at a dose rate of 0.24 Gy/min using a γ -ray source (^{60}Co , Chisostat, Chirana, Prague, Czech Republic). Single doses of 7.50 and 7.75 Gy were used.

Hematological Techniques

Concentrations of G-CSF in mouse serum were determined using an ELISA kit (R&D Systems, Minneapolis, MN). The assay had the sensitivity of 4.5 pg/ml. The animals were anesthetized intraperitoneally with Narcamon/Rometar solution [5% Narcamon and 2% Rometar (both Spofa, Czech Republic) at a ratio of 2.63:1], and the peripheral blood was sampled by cardiac puncture. For the blood cell counts, the peripheral blood was sampled from the ocular sinus. The femurs were removed and marrow cells were harvested by standard procedures. The numbers of peripheral total leukocytes and erythrocytes and the numbers of nucleated cells of the femoral marrow were determined using a Coulter Counter (Model ZF, Coulter Electronics, UK). The numbers of neutrophils were determined by evaluating peripheral blood smears stained by the May-Grünwald-Giemsa method. Standard procedures were used for the *in vitro* assays of the femoral clonogenic cells. Granulocyte-macrophage colony-forming cells (GM-CFC) were assayed using a semisolid plasma clot technique. Erythroid progenitor cells (burst-forming unit, BFU-E) were cultured on methylcellulose. Femoral marrow cell suspensions were plated (1.5×10^5 and 1×10^5 nucleated bone marrow cells for GM-CFC and BFU-E, respectively) in triplicate for both assays and incubated at 37°C in a humidified atmosphere containing 95% air and 5% CO_2 . GM-CFC were scored after a 7-day incubation as colonies containing 50 or more cells. Hemoglobinized colonies were counted as BFU-E after an 8-day incubation. For evaluation of the compartment of morphologically recognizable granulocytic and erythroid precursor cells, differential

counts were performed on marrow smears stained by the May-Grünwald-Giemsa method. In the granulocytic lineage, myeloblasts through myelocytes were designated as proliferative cells, and metamyelocytes through segmented stages as non-proliferative cells. In the erythroid lineage proerythroblasts through basophilic erythroblasts were classified as proliferative cells and polychromatic and orthochromatic erythroblasts as non-proliferative cells. The total counts of granulocytic or erythroid cells represent the sum of proliferative and non-proliferative cells, respectively.

Survival Experiments

Mice were irradiated with the lethal doses described above, and the survival was recorded daily up to day 30 after irradiation.

Statistics

Results are given as means $\pm$ SEM. For statistical comparisons of the differences between the groups, the Mann-Whitney U test was used. For assessment of survival data, the Peto & Peto Wilcoxon test was applied. The significance level was set at $P < 0.05$.

RESULTS

Survival of Mice Exposed to Mid-lethal Radiation Doses

Thirty-day survival was evaluated in two series of experiments using single radiation doses of 7.50 and 7.75 Gy γ rays. Meloxicam or saline was administered as single doses given 1 h before irradiation. After the dose of 7.50 Gy, the survival of meloxicam-treated mice ($n = 48$) was 85% compared to 64% in saline-treated controls ($n = 50$, $P < 0.01$) (Fig. 1). After the dose of 7.75 Gy, the 30-day survival in the group treated with meloxicam ($n = 20$) was

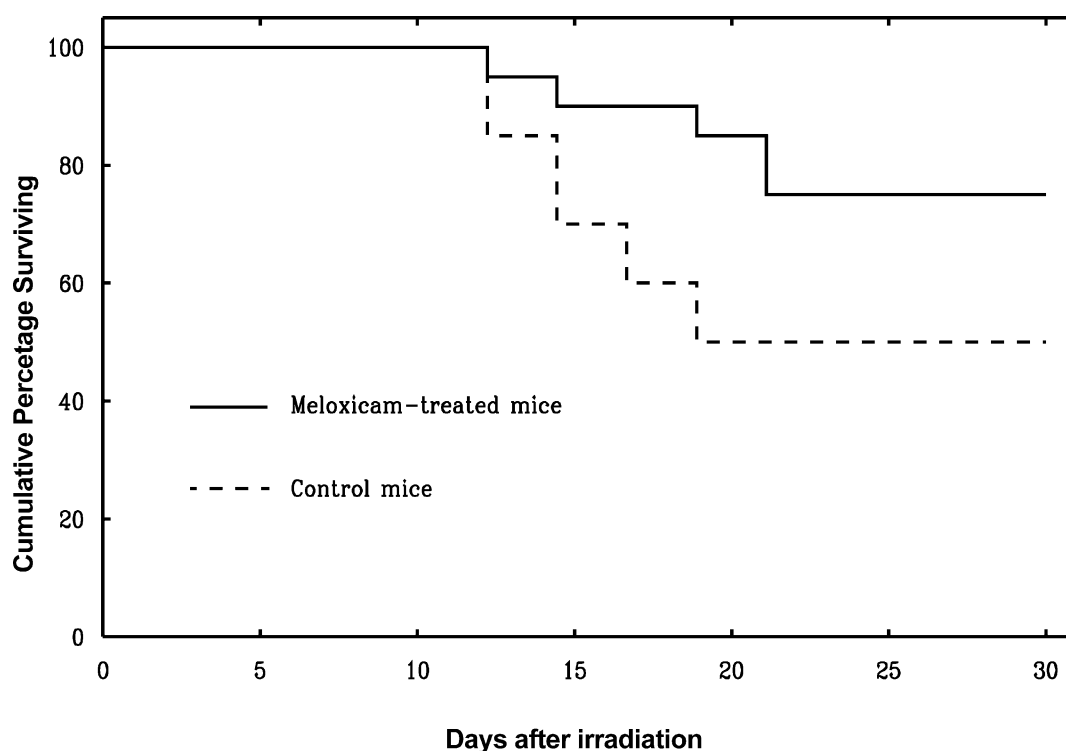


FIG. 2. Mice were given a single dose of meloxicam 1 h before irradiation with 7.75 Gy γ rays. Meloxicam-treated mice, $n = 20$; saline-treated mice, $n = 20$. Statistical significance: meloxicam-treated mice compared to saline-treated mice, $P < 0.05$ (Peto & Peto Wilcoxon test).

75% compared to 50% in saline-treated mice ($n = 20$, $P < 0.05$) (Fig. 2).

Parameters of Granulopoiesis and Erythropoiesis in Mice Exposed to a Mid-lethal Radiation Dose

In mice given a single dose of meloxicam 1 h before irradiation with a mid-lethal dose of 7.50 Gy, parameters of granulopoiesis and erythropoiesis in the femoral bone marrow and the blood were determined on day 10 after irradiation, i.e., at a time just preceding that at which the first deaths were recorded in the survival experiments. The results are summarized in Table 1. We found that hematopoiesis was strongly suppressed by radiation. Only granu-

lopoiesis was significantly influenced by meloxicam. A statistically significant increase was observed in meloxicam-treated mice in peripheral granulocyte counts as well as in the numbers of proliferative and total granulocytic cells per femur, which were 464%, 429% and 260%, respectively. The numbers of femoral GM-CFC in meloxicam-treated mice were also increased to 156%; however, this increase was not significant.

Effects of Meloxicam on Serum Levels of G-CSF

The results of experiments investigating the serum levels of G-CSF after administration of meloxicam to normal unirradiated mice are given in Table 2. Blood was sampled 6,

TABLE 1
Parameters of Granulopoiesis and Erythropoiesis in Mice Irradiated with 7.50 Gy

Parameter	Unirradiated untreated controls	Irradiated saline-treated controls	Irradiated meloxicam-treated mice
Granulocytes per 1 μ l of peripheral blood	1,130 $\pm$ 45	11 $\pm$ 11	161 $\pm$ 43**
GM-CFC per femur	17,983 $\pm$ 820	611 $\pm$ 246	953 $\pm$ 310
Proliferative granulocytic cells per femur ($\times 10^3$)	2,134 $\pm$ 253	252 $\pm$ 59	1,082 $\pm$ 184**
Total granulocytic cells per femur ($\times 10^3$)	11,198 $\pm$ 534	997 $\pm$ 250	2,591 $\pm$ 543*
Erythrocytes per 1 μ l of peripheral blood ($\times 10^6$)	8.1 $\pm$ 0.24	5.5 $\pm$ 0.4	4.8 $\pm$ 0.4
BFU-E per femur	23,736 $\pm$ 1,580	non-determinable	non-determinable
Proliferative erythroid cells per femur ($\times 10^3$)	1,807 $\pm$ 216	307 $\pm$ 53	320 $\pm$ 55
Total erythroid cells per femur ($\times 10^3$)	6,232 $\pm$ 653	1,788 $\pm$ 273	1,806 $\pm$ 319

Notes. Mice were administered meloxicam or saline in a single dose 1 h before irradiation. Parameters of granulopoiesis and erythropoiesis were determined on day 10 after irradiation. Unirradiated untreated control data are historical control data taken from ref. (16). Values are given as means $\pm$ SEM. Ten mice per group were used. *, ** $P < 0.05$, $P < 0.01$ compared to irradiated saline-treated controls (Mann-Whitney U test).

TABLE 2
Serum Concentrations of G-CSF (pg/ml) in
Unirradiated Mice

Untreated controls		
	131 ± 50	
Time after injection (h)	Saline-treated controls	Meloxicam-treated mice
6	621 ± 53*	1267 ± 63 [#]
12	208 ± 46*	1329 ± 30 [#]
24	167 ± 57	1205 ± 94 [#]

Notes. Mice were administered saline or meloxicam in a single dose. Serum concentrations of G-CSF are given as means ± SEM. Five animals per group were used. * $P < 0.05$ compared to untreated controls; [#] $P < 0.05$ compared to saline-treated controls (Mann-Whitney U test).

12 and 24 h after a single injection of the drug or saline. The serum levels of G-CSF were rather sensitive to non-specific stimuli like handling and saline injection and were increased compared to those observed in noninjected controls. After the administration of meloxicam, the serum levels of G-CSF increased further to 204%, 638% and 721% at 6, 12 and 24 h, respectively, compared to those in the respective saline-treated controls. All these differences were found to be statistically significant.

Table 3 summarizes the effects of a similar experiment performed in mice irradiated with a dose of 7.5 Gy 1 h after meloxicam administration. As in the preceding experiment, mice exposed to control handling (including irradiation) exhibit significant increases in the levels of serum G-CSF. These effects can be ascribed to some unspecified stress mechanisms. The effects of meloxicam on the levels of serum G-CSF under these conditions are similar to those observed in unirradiated mice, and the absolute values are nearly identical. At 6, 12 and 24 h after administration of meloxicam, the serum levels of G-CSF increased to 250%, 195% and 301%, respectively, of those in saline-treated irradiated mice. All these differences were found to be statistically significant.

DISCUSSION

The results shown here demonstrating increased 30-day survival of mid-lethally irradiated and meloxicam-pretreated mice are the first convincing evidence of the radioprotective potential of NSAIDs against whole-body γ irradiation. The radioprotective effects of NSAIDs described up to now have been related to the modulation of various hematopoietic functions and their postirradiation recovery after sublethal irradiation. When using classical NSAIDs, which are nonselective inhibitors of COX1 and COX2, the positive effects of the drugs on hematopoiesis can be overshadowed by undesirable side effects, and an aggravation of radiation damage can appear (see the Introduction). For these reasons, the investigations of the radioprotective po-

TABLE 3
Serum Concentrations of G-CSF (pg/ml) in Mice
Irradiated with 7.50 Gy

Unirradiated untreated controls		
	111 ± 25	
Irradiated mice		
Time after injection (h)	Saline-treated controls	Meloxicam-treated mice
6	537 ± 166*	1340 ± 22 [#]
12	638 ± 62*	1241 ± 64 [#]
24	411 ± 28*	1238 ± 121 [#]

Notes. Mice were given saline or meloxicam in a single dose 1 h before irradiation. Serum concentrations of G-CSF were determined 6, 12 and 24 h after the injection, i.e., 5, 11 and 23 h after irradiation. Values are given as means ± SEM. Five animals per group were used. * $P < 0.05$ compared to unirradiated untreated controls; [#] $P < 0.05$ compared to irradiated saline-treated mice (Mann-Whitney U test).

tential of NSAIDs that selectively inhibit COX2 represent a promising research area.

A further interesting and novel finding in our experiments is the evidence that administration of meloxicam to mice increases serum levels of the important hematopoietic cytokine G-CSF, which may activate mechanisms of the positive myelopoietic control. Thus the previous interpretation of the stimulatory effects of NSAIDs on myelopoiesis may need to be revised. Such an interpretation is based only on the negative regulatory role of prostaglandins of the E-series and thus, in the case of the inhibition of prostaglandin generation by NSAIDs, merely on the relative preponderance of the positive control (10–12). Our findings offer support for this concept by showing activation of the positive control of myelopoiesis as seen in the production of G-CSF. Although the mechanisms of the possible relationship between prostaglandin formation and G-CSF production remain to be elucidated, it is probable that the elevation of G-CSF in the serum observed in our experiments enhances the repopulation potential of the surviving hematopoietic cells and is responsible for the higher myelopoietic recovery observed on day 10 after irradiation of mice with the mid-lethal dose and for the consequent higher 30-day survival of the animals.

Before discussing both the manifestations and the mechanisms of the radioprotective action of meloxicam, it is important to mention some additional findings from the experiments. First, serum levels of G-CSF are elevated in mice treated i.p. with saline injections. This phenomenon can be explained by the function of glucocorticoids in stress. It is known that the mechanical stimulus caused by injecting the mice induces increased secretion of corticosterone (17), which, besides other actions, blocks the release of arachidonic acid from cellular phospholipids by inhibiting the activity of phospholipase A₂ and consequently inhibits prostaglandin production (18). In analogy with the effects of NSAIDs, inhibition of prostaglandin produc-

tion by glucocorticoids can also be responsible for the release of G-CSF in the case of the nonspecific stress. These findings suggest that the links between the formation of prostaglandins and the production of G-CSF can have a more general physiological meaning in the mechanisms of adaptation.

The second note relates to the action of radiation on G-CSF serum levels. Our experiments were not designed to evaluate the modulation of serum G-CSF levels by radiation. Nevertheless, the ability of glucocorticoids liberated during a non-specific stress to modulate prostaglandin production discussed above allows us to anticipate that radiation could stimulate the production of G-CSF. This should not be surprising because activation of adrenal cortex has been reported in whole-body-irradiated rodents (19). Thus it may be hypothesized that the radiation-induced stress reaction of the organism may be responsible for the enhanced serum levels of G-CSF in irradiated saline-treated controls compared to unirradiated saline-treated controls at 12 and 24 h after injection, as shown in Tables 3 and 2, respectively. These two pairs of values were compared (using the Mann-Whitney *U* test) and have been found to be significantly different ($P < 0.05$).

A characteristic feature of the protective action of meloxicam is its dependence on the radiation dose in such a way that the protective effects, manifested as increased survival, occur mainly at lower lethal doses. This is related not only to the effects of meloxicam that are probably mediated by the induction of endogenous G-CSF and manifested by enhancement of the survival of mid-lethally irradiated mice in the presented experiments and to the lack of protection at near-absolute lethal doses reported earlier (16) but also to the experiments of Tanikawa *et al.* (20), who investigated the curative effects of the repeated post-irradiation administration of exogenous rhG-CSF. Such dependence of the protective action on the radiation dose appears to be typical of the so-called biological response modifiers, i.e., means or drugs that positively stimulate regeneration processes and whose effectiveness in the case of hematopoietic damage is dependent on the presence of a certain amount of surviving stem cells (21). At higher lethal radiation doses, a disparity can arise between the activation of regenerative mechanisms and the regenerative potential of the target cell system, and thus the protective intervention may fail. Such protective mechanisms must be distinguished from those of the chemical protection induced for example by thiols that influence the development of the initial radiation damage.

With regard to the situation in which the postirradiation elevation of serum G-CSF is sufficient to induce radioprotective effects, the experiments of Sureda *et al.* (22) should be mentioned. These authors reported that a single very high dose of rhG-CSF (1 mg/kg) administered to mice 2 h after irradiation with a lethal (95%) dose significantly increased their 30-day survival (22). Such a dose of G-CSF induces a very high enhancement of serum levels of

G-CSF (to about 35 $\mu\text{g/ml}$ in 30 min after injection) in irradiated mice (23). In discussing the mechanisms of such protective effects, the authors (23) did not acknowledge the possibility that G-CSF increases production of other hematopoietic cytokines such as IL3, IL6 and GM-CSF and instead discussed only the possibility that G-CSF stimulated proliferation in the surviving pool of primitive hematopoietic precursor cells and the ability of G-CSF to decrease radiation-induced apoptosis of these cells. Similar mechanisms could also have been activated in our experiments, although at a lower intensity, as indicated by a comparison of the serum G-CSF levels we found with those obtained in their experiments (23). However, the different pharmacokinetics of G-CSF in the blood after its external administration and under conditions of its endogenous production should also be considered. While after the external administration of G-CSF its highest level in the blood peaks quickly and declines thereafter, stimulation of endogenous production of G-CSF can maintain higher levels of G-CSF in the blood for a longer time, i.e., for at least 24 h in our experimental setting. In context of these findings it is worth mentioning that the effects of G-CSF may not depend directly on its concentration but rather on the time of action of G-CSF above some particular concentration (24). For these reasons, it is difficult to relate serum levels of G-CSF to their hematopoietic efficiency.

The results of the present study suggest the possibility to use meloxicam, a selective inhibitor of COX2, in radioprotection. They also contribute to the understanding of mechanisms of the protective action of this drug. It should be noted that the radiation doses tested produce only the bone marrow syndrome. Further data are needed on the effects of meloxicam in mice irradiated with high lethal doses that induce the gastrointestinal radiation syndrome. These data would help to determine whether the effects of meloxicam on gastrointestinal toxicity make it possible to use it in the treatment of the acute radiation disease.

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REFERENCES

1. T. C. Pellmar, S. Rockwell and the Radiological/Nuclear Threat Countermeasures Working Group, Priority list of research areas for radiological nuclear threat countermeasures. *Radiat. Res.* **163**, 115–123 (2005).
2. A. D. Augustine, T. Gondré-Lewis, W. McBride, L. Miller, T. C. Pellmar and S. Rockwell, Animal models for radiation injury, protection and therapy. *Radiat. Res.* **164**, 100–109 (2005).
3. M. Pospíšil, J. Netíková and A. Kozubík, Enhancement of hematopoietic recovery by indomethacin after sublethal whole-body gamma irradiation. *Acta Radiol. Oncol.* **25**, 195–198 (1986).

4. B. A. Serushago, K. Tanaka, Y. Koga, K. Taniguchi and K. Nomoto, Positive effects of indomethacin on restoration of splenic nucleated cell populations in mice given sublethal irradiation. *Immunopharmacology* **14**, 21–26 (1987).
5. I. Nishiguchi, Y. Furuta, D. Hunter, D. Murray and L. Milas, Radioprotection of hematopoietic tissues in mice by indomethacin. *Radiat. Res.* **122**, 188–192 (1990).
6. M. Pospíšil, M. Hofer, I. Pipalová, Š. Viklická, J. Netřková and J. Šandula, Enhancement of hematopoietic recovery in gamma-irradiated mice by the joint use of diclofenac, an inhibitor of prostaglandin production, and glucan, a macrophage activator. *Exp. Hematol.* **20**, 891–895 (1992).
7. M. Hofer, M. Pospíšil, Š. Viklická, A. Vacek, I. Pipalová and A. Bartoníčková, Hematopoietic recovery in repeatedly irradiated mice can be enhanced by a repeatedly administered combination of diclofenac and glucan. *J. Leukoc. Biol.* **53**, 185–189 (1993).
8. M. Hofer, M. Pospíšil, I. Pipalová and J. Holá, Modulation of haemopoietic radiation damage by diclofenac in fractionated treatment. *Physiol. Res.* **45**, 213–220 (1996).
9. M. Hofer and M. Pospíšil, Stimulated recovery of perturbed haematopoiesis by inhibition of prostaglandin production—promising therapeutic strategy. *Cent. Eur. J. Biol.* **1**, 584–593 (2006).
10. J. Kurland and M. A. S. Moore, Modulation of hemopoiesis by prostaglandins. *Exp. Hematol.* **5**, 357–373 (1977).
11. J. Fontagné, M. Adolphe, M. Semichon, L. Zizine and P. Lechat, Effect of *in vivo* treatment with indomethacin on mouse granulocyte-macrophage colony-forming cells in culture (CFUc). Possible role of prostaglandins. *Exp. Hematol.* **8**, 1157–1164 (1980).
12. L. M. Pelus, Modulation of myelopoiesis by prostaglandin E₂: demonstration of a novel mechanism of action *in vivo*. *Immunol. Res.* **8**, 176–184 (1989).
13. G. L. Floersheim, Allopurinol, indomethacin and riboflavin enhance radiation lethality in mice. *Radiat. Res.* **139**, 240–247 (1994).
14. M. Hofer, M. Pospíšil, L. Tkadleček, Š. Viklická, I. Pipalová and J. Holá, Low survival of mice following lethal gamma-irradiation after administration of inhibitors of prostaglandin synthesis. *Physiol. Res.* **41**, 157–161 (1992).
15. G. Engelhardt, D. Homma, K. Schlegel, R. Utzmann and G. Schnitzler, Anti-inflammatory, analgesic, antipyretic and related properties of meloxicam, a new non-steroidal anti-inflammatory agent with favourable gastrointestinal tolerance. *Inflamm. Res.* **44**, 423–433 (1995).
16. M. Hofer, M. Pospíšil, V. Znojil, J. Holá, A. Vacek, L. Weiterová, D. Štreitová and A. Kozubík, Meloxicam, a cyclooxygenase 2 inhibitor, supports hematopoietic recovery in gamma-irradiated mice. *Radiat. Res.* **166**, 556–560 (2006).
17. J. Gosbee and J. Kraicer, Stress-free parenteral injections in the rat. *Can. J. Physiol. Pharmacol.* **47**, 209–210 (1969).
18. A. Munck, P. M. Guyre and N. J. Holbrook, Physiological functions of glucocorticoids in stress and their relation to pharmacological actions. *Endocrine Rev.* **5**, 25–44 (1984).
19. Z. M. Bacq and E. H. Betz, Adaptation reaction to ionizing radiation. In *Physiology and Pathology of Adaptation Mechanisms* (E. Bajusz, Ed.), pp. 185–203. Pergamon Press, Oxford, 1969.
20. S. Tanikawa, M. Nose, Y. Aoki, K. Tsuneoka, M. Shikita and N. Nara, Effects of recombinant human granulocyte colony-stimulating factor on the hematologic recovery and survival of irradiated mice. *Blood* **76**, 445–449 (1990).
21. M. L. Patchen, T. J. MacVittie and E. W. Jackson, Postirradiation glucan administration enhances the radioprotective effects of WR-2721. *Radiat. Res.* **117**, 59–69 (1989).
22. A. Sureda, A. Valls, E. Kadar, M. Algara, J. Inglés-Esteve, A. Bigas, M. Jaume, M. Lacruz, L. M. Tobajas, M. Rutllant and J. García, A single-dose of granulocyte colony-stimulating factor modifies radiation-induced death in B6D2F1 mice. *Exp. Hematol.* **21**, 1605–1607 (1993).
23. E. Kádár, A. Sureda, M. A. Mangues, J. Inglés-Esteve, A. Valls and J. García, Serum levels of G-CSF, IL-3, IL-6 and GM-CSF after a single intraperitoneal dose of rhG-CSF in lethally irradiated B₆D₂F₁ mice. *Acta Haematol.* **98**, 119–124 (1997).
24. T. Kuwabara, S. Kobayashi and Y. Sugiyama, Pharmacokinetics and pharmacodynamics of a recombinant human granulocyte colony-stimulating factor. *Drug Metab. Rev.* **28**, 625–658 (1996).